Mira Askar

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EDUCATION

University of Warwick — **Coventry, United Kingdom**
BEng Electrical and Electronic Engineering — Sep 2025 - Present
Expected graduation: July 2028 | First-year average: 74.8% (First Class) | Engineering Mathematics: 91%

Malvern College — Sep 2023 - Aug 2025
A Levels: A*AA

TECHNICAL SKILLS

Languages: Java, SQL, C/C++, MATLAB
Frameworks & Testing: Spring Boot, Hibernate ORM, Junit
Tools & Technologies: PostgreSQL, Docker, Git, REST APIs, JSON, Simulink
Hardware & Embedded: ESP32, KiCad, LTspice, CAN, UART

WORK EXPERIENCE

Healware — **Hong Kong (Remote)**
Java Backend Developer — Aug 2026 - Sep 2026

- Built Java/Spring Boot REST APIs to ingest medication selections and patient allergy histories as structured JSON, tested end-to-end with synthetic patient data.
- Designed response DTOs and API schemas returning allergy-risk assessments, contributing factors and safer medication alternatives for integration with clinical systems.
- Implemented input validation, exception handling and structured error responses with correct HTTP status codes.

Trans Asia Construction LLP — **Kyzylorda, Kazakhstan**
Java Backend Developer — Jul 2026 - Aug 2026

- Built a Java/Spring Boot backend for employee record management using Hibernate ORM and PostgreSQL, containerised with Docker.
- Designed the relational data model and CRUD workflows for employee records and document-status tracking, enabling structured search and administration.
- Implemented server-side validation, database queries and update workflows to enforce data consistency across employee records.

PROJECTS

Poker Decision Simulator
Java, JUnit, Monte Carlo Simulation

- Built a Texas Hold'em simulator in Java with full betting rounds, hand evaluation and AI-controlled opponents, using object-oriented design.
- Implemented Monte Carlo simulation to estimate hand equity for decisions under incomplete information.
- Developed an explainable fold/call/raise engine based on pot odds and expected value, covered by JUnit tests.

Self-Balancing Robot — Jun 2026
University Engineering Project | MATLAB, Simulink

- Developed Simulink control code to process sensor data and maintain balance; tuned controller coefficients in MATLAB Simulink and achieved successful balancing in practical testing.
- Achieved a First Class mark for the product, team presentation and planning.

Hyperloop University Project — Oct 2025 - Present
Electronics Hardware Engineer | KiCad, LTspice, CAN, UART

- Designed a custom battery-temperature monitoring PCB in KiCad (schematic and layout), validating circuit behaviour in LTspice before prototyping.
- Integrated CAN and UART interfaces to connect the temperature-monitoring system to the Battery Management System.

ACHIEVEMENTS & INTERESTS

Awards: British Physics Olympiad (BPhO) - Gold; Gold CREST Award for QinetiQ project.
Scholarships: University of Warwick Global Excellence Scholarship; Malvern College Honorary Scholarship.
Sponsorship: Secured approximately £1,000 from KT Lab Cloud and Chevron at the Enactus National Expo Kazakhstan.
Interests: Piano (completed music school with a final grade 5.0/5.0); choir and orchestra; volleyball at the University of Warwick; robotics.